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Method for classification of UAV flight control RF signals based on multi-scale divergence entropy and optimized neural networks

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To address the critical challenge of insufficient classification accuracy for UAV flight control radio frequency (RF) signals in complex electromagnetic environments, this study proposes a novel classification framework combining Multiscale Dispersion Entropy (MDE) feature fusion with an Artificial Lemming Algorithm (ALA)-optimized BP neural network. The proposed method first employs MDE to extract robust multiscale dynamic features from RF signals, constructing a discriminative 12-dimensional feature matrix, then utilizes the biologically inspired ALA to globally optimize the BP network's weights, biases, and hidden layer architecture through its unique migration-burrowing-foraging-predator avoidance mechanisms that dynamically balance exploration and exploitation. Extensive experiments on the DroneRFa dataset containing six mainstream UAV models demonstrate the framework's superior performance, achieving 97.2% classification accuracy (a 4.7-7.1% improvement over conventional GA-BP and PSO-BP methods), remarkable noise robustness (maintaining 90% accuracy at SNR=0 dB), and accelerated convergence (reaching 90% accuracy in just 65 iterations), with further validation from ROC analysis showing an exceptional AUC of 0.97. These results collectively confirm that the MDE-ALA-BP approach effectively overcomes the limitations of existing methods in complex electromagnetic scenarios, providing a reliable and efficient technical solution for enhanced airspace security monitoring and low-altitude traffic management systems.

Keywords Unmanned aerial vehicle, Flight control radio frequency signals, Signal classification, Multiscale dispersion entropy, Optimized neural networks

As one of the most disruptive technological innovations in the 21 st century, UAV technology has deeply penetrated the fields of military reconnaissance, logistics and transportation, precision agriculture, environmental monitoring, and disaster emergency response[1–3]. However, concomitant issues such as airspace security threats, privacy leakage risks, and illegal misuse hazards have posed serious challenges to UAV regulatory technologies. In this context, the efficient identification technology of non-cooperative UAVs has become the core technical bottleneck for safeguarding national airspace security and preventing air traffic collisions, as well as the key support for realizing the cooperative operation of multiple aircraft in low-altitude intelligent traffic management (UTM) systems[4,5].

Existing UAV detection technology systems mainly contain mainstream paradigms such as radar detection[6,7], optoelectronic/infrared sensing[8,9], acoustic recognition[10], RF signal parsing[11] and deep learning-based intelligent recognition[12].The performance of different technological approaches exhibits significant disparities. Radar technology, as a traditional detection method, can obtain high-precision spatial coordinates, but it has inherent resolution limitations for detecting Low-Slow-Small (LSS) targets and is prone to false alarms due to complex ground clutter. Optoelectronic/infrared sensors rely on target spectra to achieve UAV identification, making them particularly suitable for nighttime or covert operations. However, their detection efficiency is constrained by atmospheric attenuation effects, and the signal-to-noise ratio deteriorates significantly under adverse weather conditions such as fog and rain, drastically reducing detection performance.The acoustic detection method locates targets by capturing sound signals generated by UAVs. While this approach is low-cost and easy to deploy,

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its engineering practicality is limited by the high environmental sensitivity and short propagation distance of acoustic signals. Deep learning-based visual recognition methods demonstrate excellent performance in target classification tasks but suffer from inherent drawbacks such as insufficient model generalization capability, vulnerability to adversarial samples, and heavy reliance on labeled data. Additionally, they struggle to adapt to new or unknown UAV models.

RF signal detection technology realizes target identification by parsing the communication signal between the UAV and the controller, showing unique technical advantages: firstly, the RF signal has the ability of non-line-of-sight (NLOS) penetration, which can effectively overcome the physical limitations of the traditional optical sensors such as building obstruction and complex terrain; Secondly, the technology has all-weather operation stability, and its performance is not interfered with by external variables such as lighting conditions and meteorological environment, which is especially suitable for critical scenarios requiring continuous monitoring such as border security and airport perimeter;Then, RF signals can capture features specific to the UAV manipulation link, providing an essential differentiation basis for the construction of a highly robust recognition model; finally, RF signal detection and recognition also has an early warning function, which is important for the prevention of UAV-related safety accidents, as RF signals can be captured at a longer distance, and thus can provide earlier warnings. Nevertheless, the existing RF identification methods still have core problems such as single dimension of signal feature extraction and insufficient classification accuracy in complex electromagnetic environments[13,14].

This paper proposes a hierarchical recognition framework for UAV RF signals based on Multi-scale Dispersion Entropy (MDE) feature fusion and Artificial Lemming Algorithm (ALA) optimized BP neural network.The primary objective is to address the critical issue of slow convergence and low accuracy of traditional methods in complex environments. We argue that superior convergence performance and noise robustness are fundamental prerequisites for practical deployment. Validation is performed by constructing an open-source RF signal dataset containing six mainstream UAV models. The experimental results show that the proposed method has significant advantages in convergence speed, classification accuracy, and feature characterization capability, thereby providing a high-performance algorithmic foundation for non-cooperative UAV recognition applications.

Multi-scale dispersion entropy feature extraction methods

Multi-scale dispersion entropy theory

Multiscale dispersion entropy (MDE) has high robustness concerning noise, and can effectively extract useful feature information even in noisy data. It can analyze signals on multiple time scales, thus capturing the dynamics of signals at different frequency components. Compared with some other complex entropy measurements, the multi-scale scattering entropy calculation is more efficient and more practical when dealing with large-scale data sets[15,16].

For any time series of length N $X = \{x_1, x_2, ..., x_N\}$, can be reconstructed in the embedding dimension m as a new time series shown in the following equation.

$$\begin{aligned} Y(m) &= \{y_1(m), y_2(m), ..., y_{N-m+1}(m)\} \\ &= \left\{ \begin{array}{cccc} x_1 & x_2 & \cdots & x_m \\ x_2 & x_3 & \cdots & x_{m+1} \\ \vdots & \vdots & & \vdots \\ x_{N-m+1} & x_{N-m} & \cdots & x_N \end{array} \right\} \end{aligned} \tag{1}$$

The cosine similarity between neighboring reconstructed sequences is calculated and can be obtained according to the following equation:

$$\begin{aligned} D(m) &= \{d_1, ..., d_{N-m}\} \\ &= \{\ d(y_1(m), y_2(m)), d(y_2(m), y_3(m)), \\ &...,d(y_{N-m-1}(n), y_{N-m}(m))\} \end{aligned} \tag{2}$$

among others,

$$d(y_i(m), y_j(m)) = \frac{\sum_{k=1}^{m} y_i(k) \times y_j(k)}{\sqrt{\sum_{k=1}^{m} y_i(k)^2} \times \sqrt{\sum_{k=1}^{m} y_j(k)^2}} \in [-1, 1] \tag{3}$$

The size of the cosine similarity indicates the ordered state of the time series $X = \{x_1, x_2, ..., x_N\}$, the larger the cosine similarity indicates that the time series is more ordered, and the smaller the cosine similarity indicates that the time series is more diverse. Divide the interval [−1,1] of the above equation into different intervals, then calculate the probability that the cosine similarity falls into each interval, and then get the state probability set $(P_1, P_2, ..., P_\varepsilon)$, and satisfy $\sum_{k=1}^{\varepsilon} P_k = 1 (I_1, I_2, ..., I_\varepsilon)$

The Dispersion Entropy (DE) can be calculated from the set of state probabilities obtained earlier as follows:

$$DE(m,\varepsilon) = -\frac{1}{\ln \varepsilon}\sum_{k=1}^{\varepsilon} P_k \ln P_k \in [0,1] \tag{4}$$

Scatter entropy, in the physical sense, indicates that the time series is stable when DE tends to 0. When DE tends to 1, it indicates that the time series has random, chaotic and unstable phenomena.

Scatter entropy is information about the time series computed from a single scale, and multiscale analysis can extract information about the time series from different scales separately. An important step in the multiscale analysis is the coarse-graining process, by which the time series can be calculated and analyzed at multiple time scales. Firstly, the given time series $X = \{x_1, x_2, ..., x_N\}$ is decomposed into a new series $\{y_j^\tau\}$ at multiple scales, and the decomposition process is given in the following equation:

$$y_j^\tau = \frac{1}{\tau}\sum_{i=j}^{j+\tau-1} x_i,\ 1 < i < N - \tau + 1 \tag{5}$$

Where τ denotes the multiscale factor, a positive integer, and the multiscale factor aims to quantify the dynamic characteristics of the time series at different time scales. When $\tau = 1$, the multiscale sequence $\{y^{(1)}\}$ is the original time series.

The scatter entropy value is calculated for each multiscale time series, i.e., Multiscale Dispersion Entropy (MDE) is obtained. It is shown below:

$$MDE(X,\tau,m,\varepsilon) = DE(y_j^\tau, m, \varepsilon) \tag{6}$$

Traditional single-scale analysis methods cannot fully reflect all the characteristics of a signal. In contrast, multi-scale scattering entropy can analyze signals on multiple time scales, thus capturing the intrinsic characteristics and underlying patterns of signals more comprehensively[17]. Moreover, by analyzing the signal at multiple scales, the influence of noise can be reduced to a certain extent, because noise tends to exhibit different characteristics at different scales. The paper quantifies the dispersion entropy of the UAV flight control RF signals at multiple scales, which provides richer and more detailed feature representations for the subsequent pattern recognition of the signals and improves the recognition.

Data pre-processing

The DroneRFa dataset[18] contains 24 common UAV flight control RF signals, some example models include DJI Phantom 3, DJI AVATA, DJI MATRICE 200, etc., covering three open frequency bands: 915 MHz, 2.4 GHz and 5.8 GHz.The figure 1 below shows the six UAVs for which the paper was conducted.In this study, only six DJI platforms are selected as the initial verification set among the 24 UAV signals provided by DroneRFa, mainly based on the three methodological considerations of control variables, contribution quantification and reproducibility : DJI products occupy more than 70 % of the global civil market, and their RF links are

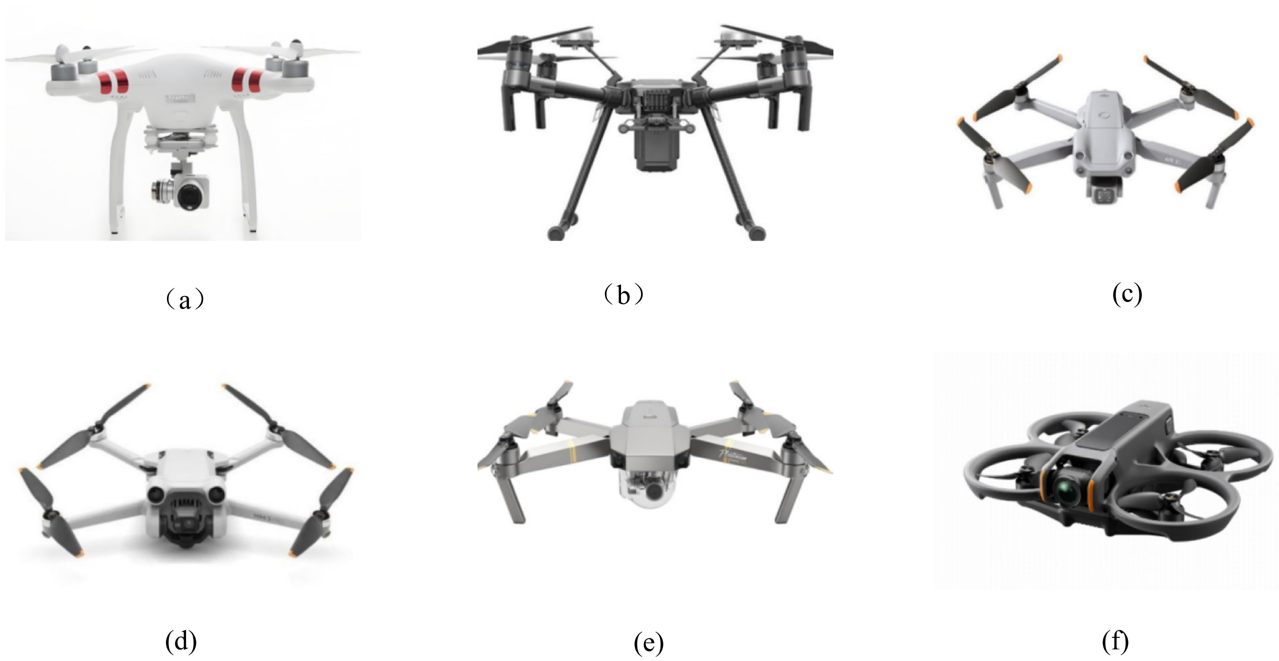

Fig. 1. Several DJI drones for experiments.

highly homologous in chip scheme, frame encapsulation format and frequency hopping strategy. Fixed brands can eliminate the inter-class variance introduced by RF front-end differences to the greatest extent. Six types of sufficient samples have met the statistical power of 5 × 2 cross-validation, and can directly quantify the convergence gain and accuracy increase of the algorithm without hardware heterogeneous noise interference), so as to avoid the dilution of contribution. And the subset I/Q sampling parameters are consistent, ensuring that the experimental conditions can be reproduced at zero cost.

The data in the dataset is divided into four channels,’RF0_I’,’RF0_Q’,’RF1_I’and ‘RF1_Q’. Wherein,’RF0’and RF1’ represent the first and second channels of the radio frequency receiver, respectively, ‘I’denotes the in-phase component,and “Q”represents the quadrature component. Due to the limitation of equipment read/write performance, the data acquisition adopts the mode of ‘pick-store-pick-store’, and each acquisition is set to 10 megabytes of data points, to adapt to the input requirements of the subsequent UAV recognition model. First, the original RF signal is stored in the form of two-channel complex numbers (I/Q components), and in order to eliminate the effects of environmental noise and equipment acquisition errors, each signal fragment is normalized to ensure that the signal amplitude distribution is in the [−1,1] interval. Then, 10,000 sampling points in each data set were intercepted for analysis, and the figure 2 below shows the intercepted four-channel RF signal of DJI Mavic Pro.

Multi-channel multi-scale dispersion entropy feature extraction method

The selection of MDE multiscale factors directly affects the significance of the differences in signal characteristics. Therefore, the paper employs the Grid Search method to systematically traverse the parameter combinations, the cross-validation method is used to evaluate their performance. The initial multi-scale factor τ is set to take values in the range of 1–20, and the search step is set to 0.5, covering the short, medium and long time scales of the UAV flight control RF signals.A 5-fold cross-validation is used. The dataset was divided into five parts, four of which were used for training and the remaining one for testing, this process was repeated five times, each time selecting a different part of the set as the test set. Each set of pairs of scale factor parameter combinations is set as a 3-dimensional matrix, with each element of the matrix being a multiscale factor, resulting in a 3-dimensional feature matrix. The 3-dimensional feature difference significance index after each training is used as the objective function, and the parameter combination with the most significant difference is finally selected as the final time-scale factor combination, and the difference significance analysis is selected to be implemented by ANOVA (Analysis of Variance), which is used to compare whether there is a significant difference in the means between three or more groups, and it analyses the source of variation in the data to determine whether the differences between groups are significantly greater than the differences within groups.A flowchart of the above process is shown in Figure 3:

The grid search for optimal scale factors used 5-fold cross-validation on the training set only to prevent data leakage. The final model was trained on the full training set (70%) and evaluated on the held-out test set (30%).After the above parameter combination search, the multiscale factor combination $[\tau_1, \tau_2, \tau_3]$ with the

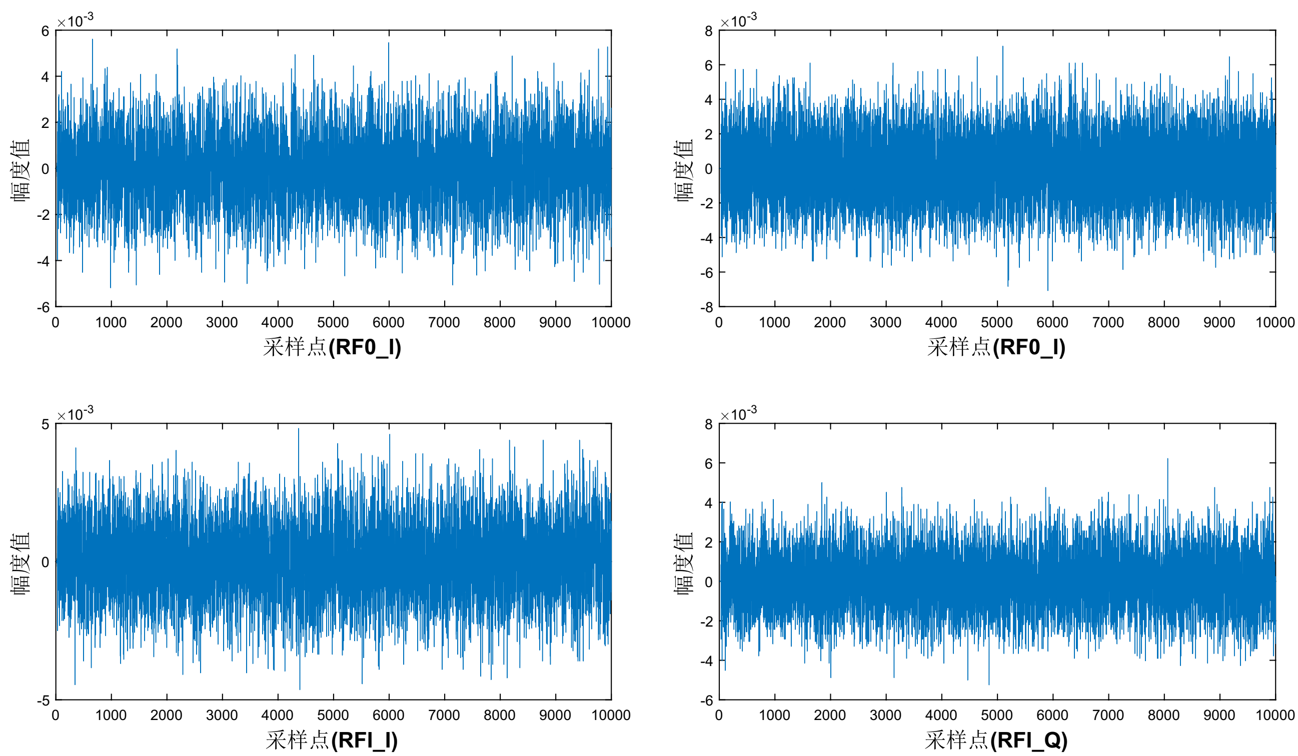

Fig. 2. DJI MavicPro UAV quad-channel flight control RF signal.

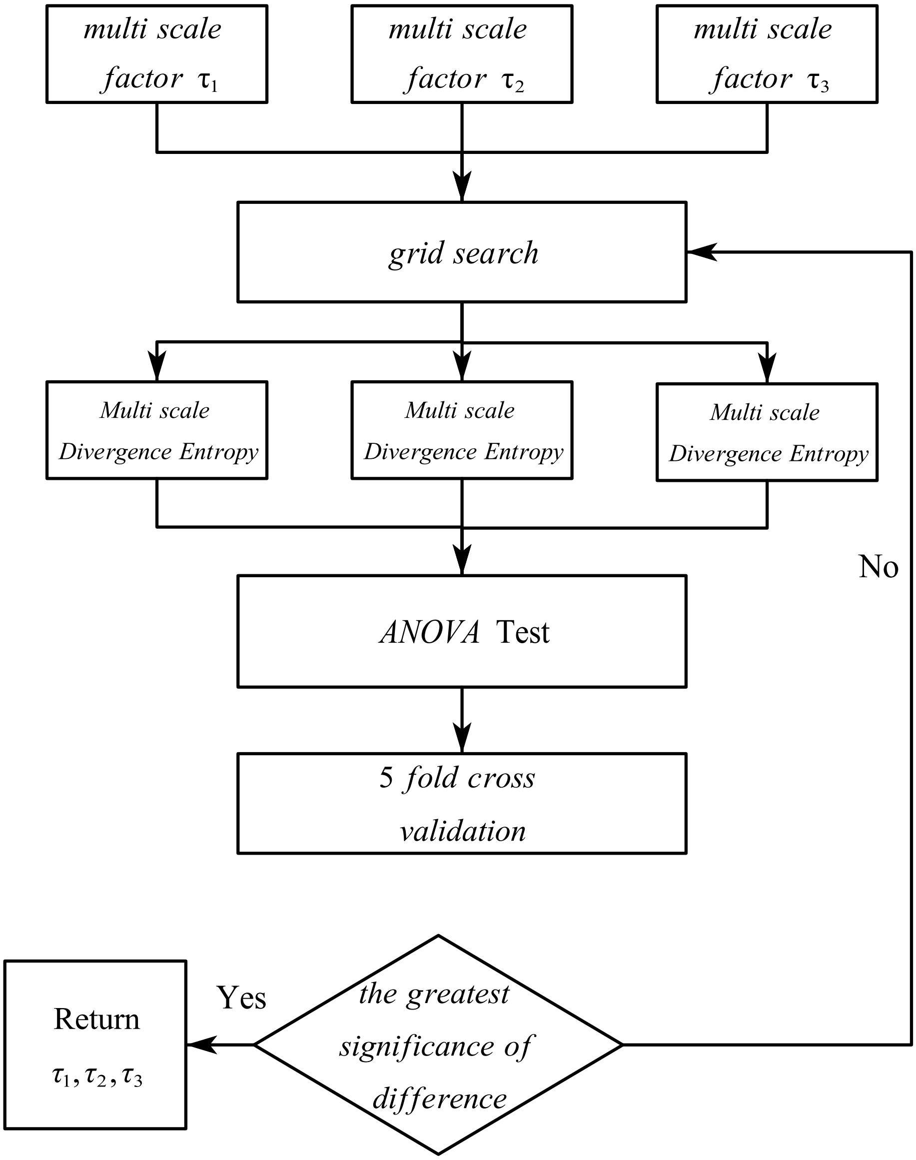

Fig. 3. Flowchart of multi-scale factor search.

most significant difference is filtered out, and the following figure 4 shows the results of the multiscale factor combination search using the grid search method, and the points pointed out by the arrows in the figure are the optimal scale factor combination points, and the $[\tau_1, \tau_2, \tau_3]$ completed by searching is substituted into the above equation (6) to get the MDEs of different multiscale factors, and the MDEs in the cases of multiscale factors τ_1,τ_2 and τ_3 are denoted as MED1, MDE2, and MDE3, respectively, which are constructed into the feature matrix $[MDE_1, MDE_2, MDE_3]$

The table 1 below shows the grid search process in the objective function that is the final test results of the ANOVA test, where the original hypothesis H0 is that the means of all groups are equal and there is no significant difference, set the significance level of 0.05, the ANOVA test returns a P-value of 0.00023 is much smaller than 0.05, so the original hypothesis can be rejected, accept the alternative hypothesis H1: the means of all groups are unequal and there is a difference in the Significance.

The flight control RF signals of six types of UAVs, namely DJI Phantom 3, MATRICE 200, Air 2S, Mini 3 Pro, Mavic Pro, and AVATA, are selected for MDE feature extraction. The following figure 5 shows the three-dimensional three-point distribution map after feature extraction of the flight control RF signals of the six selected types of UAVs, from which it can be seen that the MDE features of the different types of UAVs have certain distributional differences in the three-dimensional space, and the feature points of the different types of UAVs overlap less and are scattered and distributed.

In the DroneRFa dataset, the flight control RF data of each type of UAV includes four channels : ' RF0 _ I ', ' RF0 _ Q ', ' RF1 _ I ' and ' RF1 _ Q '. The MDE feature extraction is performed on the signals of each channel, that is, the signals of each channel can extract the three-dimensional features of MDE1, MDE2 and MDE3, and then the MDE features of different channels are fused. The four-channel data of each type of UAV can

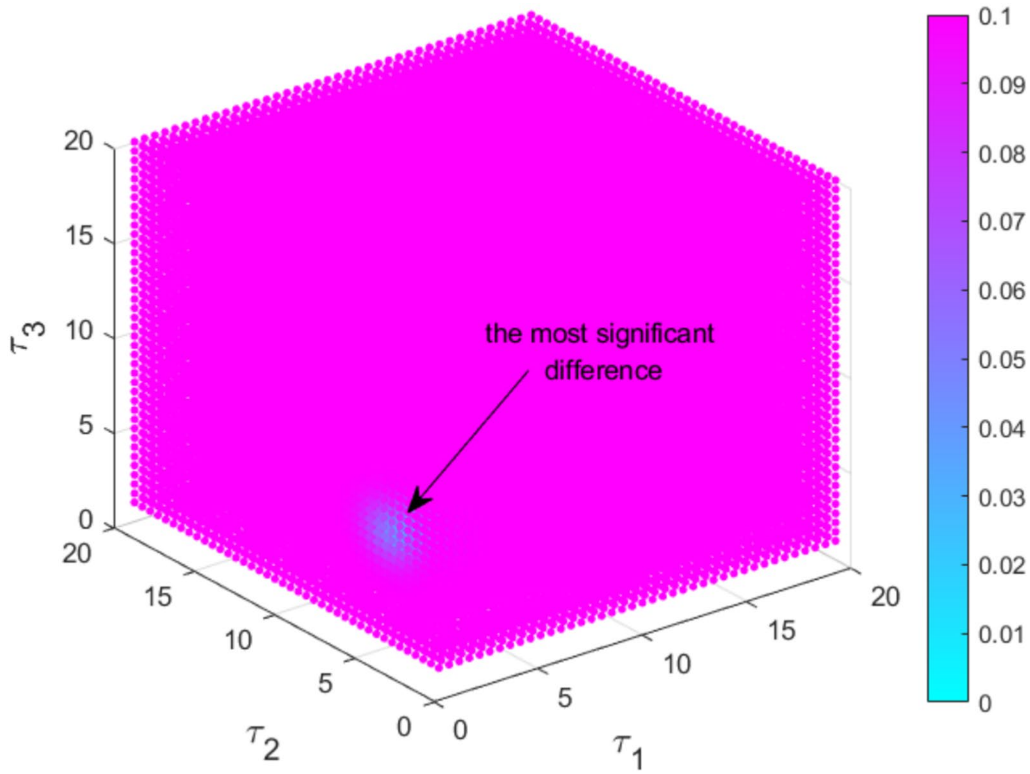

Fig. 4. Grid search method to find the combination of factors with the largest scale of difference significance.

Value of τ_1	Value of τ_2	Value of τ_3	Hypothesis H_0	Value of P	Strategic decision
2.5	4.4	6.3	Equal all means for groups	0.00023	Reject the original hypothesis and accept the alternative hypothesis

Table 1. Grid search ANOVA test results (significance level 0.05).

extract 12-dimensional features and construct a 12-dimensional feature matrix. At this point, the multi-channel multi-scale divergence entropy feature extraction of the UAV flight control RF signal is completed, and the corresponding feature matrix is constructed.The following figure 6 shows the structural schematic diagram of the multi-channel pair-scale divergence entropy feature matrix constructed. From the diagram, it can be seen that the four-channel data of each type of UAV flight control RF signal can be used for MDE feature extraction. The MDE features of three scale factors are extracted from each channel data, and finally, a feature matrix of four rows and three columns can be formed. The feature matrix is reconstructed into a one-dimensional row matrix as the input of the neural network in subsequent recognition.

Optimisation neural network based RF signal classification method for UAV flight control

Traditional BP neural network algorithms are based on the gradient descent method of adjusting the weights, which can easily cause the optimization process to fall into local minima rather than global minima. This is especially common in complex, high-dimensional spaces. Moreover, when the amount of input data is large or the network structure is complex, the BP neural network algorithm takes a long time to converge to a solution. This is due to the fact that each weight update is based only on reducing the current error, rather than moving directly towards an optimal solution.The final performance relies heavily on the choice of initial weights, and inappropriate initial weights may lead to training failures or require more iterations to achieve better performance[19].

Intelligent optimization algorithms have global search capability, which can explore the solution space more effectively and find better weights and bias values, thus avoiding the problem of falling into local minima. The intelligent optimization algorithm can automatically adjust its own parameters, inertia weights, learning factors, etc., according to the feedback information in the optimization process, which helps to improve the optimization efficiency and stability.It is worth noting that the field of nature-inspired metaheuristic algorithms is rapidly evolving, with recent introductions of novel algorithms such as the Hippopotamus Optimization Algorithm[20] and others[21–24], further enriching the toolbox for solving complex engineering problems.

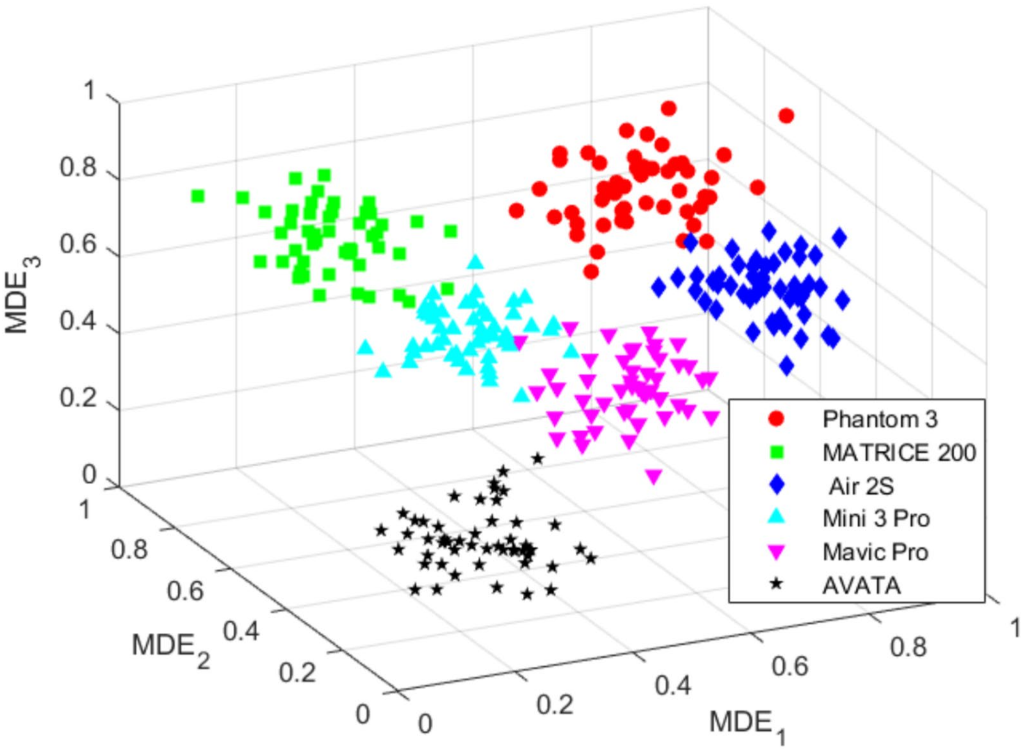

Fig. 5. Scatter plot of MDE features of flight control RF signals of different UAV models.

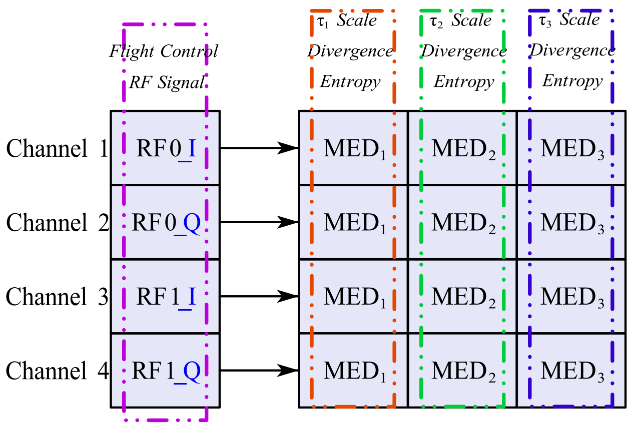

Fig. 6. Schematic structure of four-channel multi-scale dispersion entropy feature matrix.

Principles of the artificial lemming algorithm

The Artificial Lemming Algorithm (ALA) is a meta-heuristic algorithm inspired by four natural behaviors of lemming[25]: long-distance migration (exploration): simulating lemming group migration in search of new habitats; burrowing behavior (exploration): simulating lemmings digging complex tunnel systems; foraging behavior (exploring): simulating lemmings randomly searching for food in a local area; and predator evasion (exploring): simulating lemmings' escape strategy of quickly escaping back to their burrows. The exploration and exploitation are dynamically balanced by the energy-decreasing mechanism, focusing on global exploration in the early stage and local exploitation in the later stage.The main algorithm steps are as follows:

Initialisation

In the initialisation phase, ALA initiates its optimisation process by generating a random population in the search space as shown in the following equation:

$$z_{i,j} = LB_j + rand \times (UB_j - LB_j),\ i = 1, 2, ..., N,\ j = 1, 2, ..., Dim \tag{7}$$

Migration phase
Lemmings migrate randomly over long distances in search of more suitable habitats. In this process, lemmings explore their surroundings based on their current location and the location of other random individuals in the group, aiming to discover areas with more abundant food resources, thus improving their survival and access to more resources. The behavior is represented as follows:

$$\vec{Z}_i(t+1) = \vec{Z}_{best}(t) + F \times \overrightarrow{BM} \times (\vec{R} \times (\vec{Z}_{best}(t) - \vec{Z}_i(t)) + (1-\vec{R}) \times (\vec{Z}_i(t) - \vec{Z}_a(t))) \tag{8}$$

where $\vec{Z}_i(t)$ denotes the position of the i search agent at t+1 iterations. Where, $\vec{Z}_{best}(t)$ denotes the current optimal solution.

$$F = \begin{cases} 1, if \lfloor 2 \times rand + 1 \rfloor = 1 \\ -1, if \lfloor 2 \times rand + 1 \rfloor = 2 \end{cases}, \vec{R} = 2 \times rand(1, Dim) - 1$$

Nesting stage
Lemmings will randomly dig new burrows based on the current location of the burrow and the location of random individuals in the population. It helps them to escape quickly from the threat of predators and find food more efficiently. The process is represented as follows:

$$\vec{Z}_i(t+1) = \vec{Z}_i(t) + F \times L \times (\vec{Z}_{best}(t) - \vec{Z}_b(t)) \tag{9}$$

Where L is a random number associated with the current iteration number. Where $\vec{Z}_b$ denotes a randomly selected search individual from the population and b is a random integer indicator value between 1 and n. L and $\vec{Z}_b$ are used to describe the interactions of the lemming individuals while digging a new hole.

Foraging phase
The third behavioural pattern of lemmings is to move randomly through the burrows of their habitat in search of food sites.Searching for food. Lemmings usually define a relatively small foraging area within their living area, and in order to be able to consume the maximum amount of food, lemmings will roam freely within that foraging area without a fixed path. The process is as follows

$$\vec{Z}_i(t+1) = \vec{Z}_{best}(t) + F \times rand \times \vec{Z}_i(t) \tag{10}$$

Avoiding natural enemies
This stage focuses on the lemming's evasive behaviour in the face of a threat, as soon as a threat is perceived, the lemming will return to its burrow to take refuge. At the same time, the lemming performs disorienting manoeuvres to confuse the opponent. The process is illustrated below:

$$\vec{Z}_i(t+1) = \vec{Z}_{best}(t) + F \times G \times (\vec{Z}_{best}(t) - \vec{Z}_i(t)) \tag{11}$$

Transition phase
In order to maintain a balance between the above processes, the energy factor decreases during the iterations. When lemmings have enough energy, they selectively migrate or dig holes; otherwise, they forage around to avoid predators. The process is represented as follows:

$$E(t) = 4 \times \arctan\left[1 - \frac{t}{T_{\max}}\right] \times \ln(\frac{1}{rand}) \tag{12}$$

ALA-BP neural network optimisation scheme

Problem modelling
In the ALA-BP neural network optimization scheme, the problem of optimizing the weights, bias and the number of nodes in the hidden layer of a BP neural network is transformed into a multidimensional spatial search problem.

The weights and biases of the neural network are expanded into a one-dimensional vector that serves as the search agent location for ALA:

$$\vec{Z}_i = [w_{11}^{(1)}, w_{12}^{(1)}, ..., w_{nm}^{(L)}, b_1^{(1)}, ..., b_m^{(L)}, N_1^{(1)}, ..., N_n^{(L)}] \tag{13}$$

Where $W_{nm}^{(L)} \in R^{n \times m}$ is the weight matrix, $b_m^{(L)} \in R^m$ is the bias vector and $N_n^{(L)} \in R^n$ is the number of hidden layer nodes.

The fitness function is set to:

$$Fitness = \frac{1}{N}\sum_{i=1}^{N}(y_{pred}^{(i)} - y_{true}^{(i)})^2 \tag{14}$$

Process optimisation
The optimal calculation of BP neural network parameters is carried out by ALA according to the flow chart below (Fig. 7):

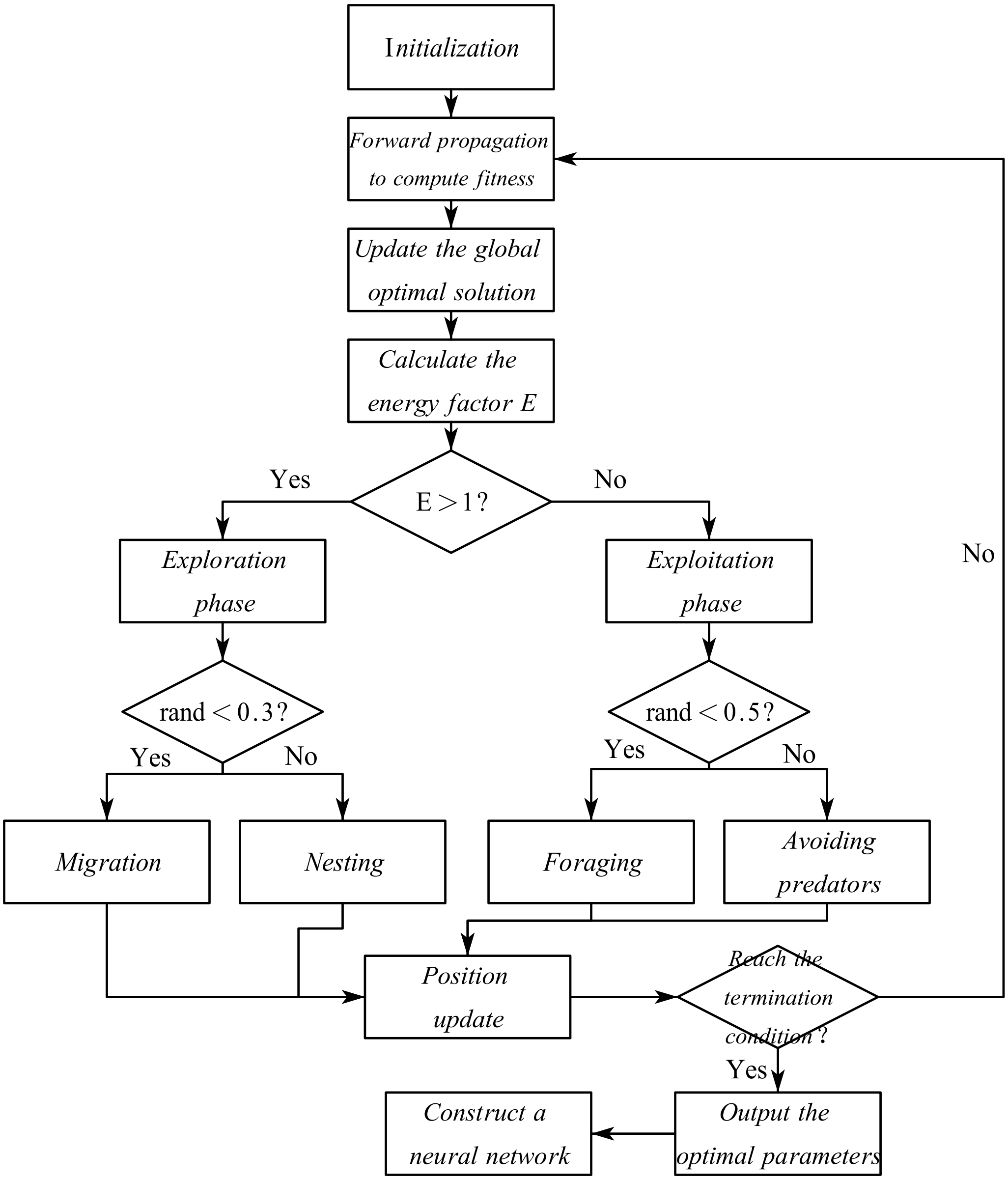

Fig. 7. Flowchart of ALA-BP neural network algorithm.

The pseudo-code representation of the algorithm shown above is as follows.

Inputs: training dataset TrainData**,** testing dataset TestData, population size P, maximum number of iterations IterMax

Output: optimised BP neural network model

1. Initialise the population Population, each individual represents a parameter of the BP neural network

2. Iterate IterMax times:

a. For each individual in Population:

i. Migration: update individual locations based on the current optimal solution and other individual locations

ii. Nesting: digging new caves and updating locations based on cave locations and randomised individual locations

iii. Foraging: random searches in localised areas, updating positions

iv. Hiding from predators: judge the threat, return to the cave for shelter, update position

b. Transition: reducing the energy factor and balancing exploration and exploitation

c. Calculate the fitness value (classification error) for each individual

d. Retain the individual with the highest fitness as the current optimal solution

3.Configuration of the BP neural network using the parameters of the optimal individual.

4.Evaluation of model performance on test data sets

5.Return the optimised BP neural network model

Optimisation neural network based RF signal classification method for UAV flight control

Algorithm integration and network structure design

The BP neural network employed in this study is a standard single-hidden-layer Multilayer Perceptron (MLP) trained using backpropagation. The input layer consists of 12 nodes, corresponding to the 12-dimensional MDE feature vector extracted from the quad-channel RF signals. The output layer comprises 6 nodes, each representing one of the six UAV classes, with a Softmax activation function to produce class probability distributions. The number of hidden neurons is dynamically optimized by the Artificial Lemming Algorithm (ALA) within the range of 5 to 20. This range was selected based on preliminary experiments and cross-validation, aiming to balance model capacity and generalization: values below 5 were found insufficient to capture complex patterns, while values above 20 led to overfitting due to the relatively small dataset size. The hidden layer uses the ReLU activation function to enhance nonlinearity and accelerate convergence. The loss function is categorical cross-entropy. Input features are normalized to the interval $[-1, 1]$ using min-max scaling. No additional regularization techniques (such as dropout or L2 weight decay) are applied, as the global optimization nature of ALA and the controlled experimental setup inherently reduce overfitting risks. Training employs early stopping with a patience of 30 epochs, monitoring the validation loss to prevent overfitting.

Based on the multi-channel Multi-scale Dispersion Entropy (MDE) feature matrix (12-dimensional features) extracted in the previous section, a BP neural network classification model (ALA-BP) optimized based on the artificial lemming algorithm (ALA) is proposed. The model optimizes the weights, biases and the number of hidden layer nodes of the neural network through the global search capability of ALA, so as to improve the classification performance. The network structure is designed according to the following process:

(1) Input layer: receives 12-dimensional MDE feature vector with 12 nodes.
(2) Hidden layer: a single hidden layer structure is used, and the initial number of nodes is determined by ALA optimisation (search range: 5–20). The hidden layer activation function selects ReLU (Rectified Linear Unit) kernel function to enhance the nonlinear expression ability.
(3) Output layer: the number of nodes corresponds to the number of UAV types (6 categories), and the Softmax function is used to output the probability distribution.

The optimization variables are chosen to be a matrix consisting of a weight matrix $\vec{Z}_i$, bias vector and a number of hidden layer nodes, and the fitness function is optimized with the classification mean square error as the optimization objective, as shown in equation (14) above. Constraints: the network training error needs to converge to the threshold ($< 10^{-4}$) to prevent overfitting.

The dataset was divided using stratified sampling to divide the DroneRFa dataset into training and testing sets in 7:3 to ensure a balanced proportion of each category. The 12-dimensional MDE features were Z-score normalized to eliminate the effect of magnitude. Set the population size $N = 30$,the maximum number of iterations $T\_\max = 300$, and the search space dimension $Dim = L + m + n$. The learning rate of the neural network was adaptively adjusted with an initial value of 0.01 and a decay rate of 0.99/epoch, and an early stop mechanism was set to terminate the training if the loss of the training set did not decrease for 10 consecutive times.A graph of the training loss with the number of iterations is shown in the figure below, from which it can be seen that the training loss exhibits a gradually decreasing trend, which suggests that the adopted neural network model is able to effectively learn from the data and gradually optimize its parameters in order to reduce the prediction error. However, more significant fluctuations in the error values can be observed in the early stages of training, which may be attributed to the local minima or saddle points encountered by the model while exploring the data feature space, and these fluctuations imply that the model is more sensitive to the initial weight settings and input data in the early stages.

The feature input process inputs the preprocessed MDE feature vectors into the ALA-BP network, and obtains the confidence level of each category through the nonlinear transformation of the hidden layer and the calculation of the probability of the output layer, and selects the category with the highest confidence level as the final classification result.Figure 8 shows the curve of training loss with the number of iterations.

Comparative experiments on classification recognition accuracy

To verify the superiority of ALA-BP, the following control group was set up:

(1) Conventional BP network: fixed number of hidden layer nodes (10), using random initial weights;
(2) Genetic Algorithm Optimisation BP (GA-BP): population size 50, crossover probability 0.8, variance probability 0. 1;
(3) Particle Swarm Optimisation BP (PSO-BP): number of particles 50, inertia weight 0.6, learning factor $c_1 = c_2 = 1.5$;
(4) Grey wolf optimised BP (GWO-BP): population size and dimensions as in ALA-B.

Two evaluation metrics, Accuracy and ROC curve (Receiver Operating Characteristic Curve) are selected. The following figures show the final results of the comparison experiments. Among them, Figure 9 shows the results of the comparison experiment for the classification accuracy metric.

The effectiveness of the proposed ALA-BP model is evaluated through a comparative analysis with traditional BP, GA-BP, PSO-BP, and GWO-BP algorithms. The theoretical basis lies in the ability of intelligent optimization algorithms to explore the solution space more efficiently and find better weights and bias values. Specifically, ALA-BP is able to find globally optimal solutions in complex, high-dimensional spaces by dynamically balancing the process of exploration and exploitation. Experimental results show that ALA-BP outperforms other algorithms in terms of convergence speed, final accuracy and stability, which is mainly attributed to its global search capability and insensitivity to initial weights.

ROC Comparison experiment

In order to show the performance differences of the algorithms more intuitively,table 2 shows a comparison table of curve features, which summarizes the performance of different algorithms in terms of convergence speed, final accuracy and range of fluctuation in the late stage, etc. ALA-BP exhibits the fastest convergence speed, and it only needs about 65 iterations to reach 90% classification accuracy, which is significantly better than other algorithms. After 200 iterations, the classification accuracy of ALA-BP is stable at 97.2%±0.3%, showing extremely high classification accuracy and stability. 0.3%, showing extremely high classification accuracy and

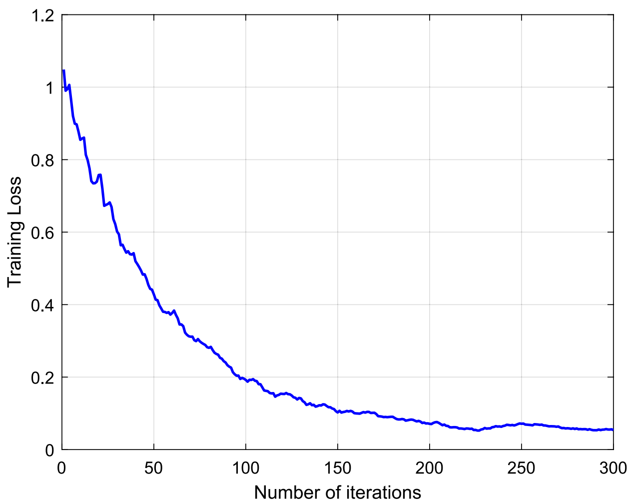

Fig. 8. Curve of training loss with the number of iterations.

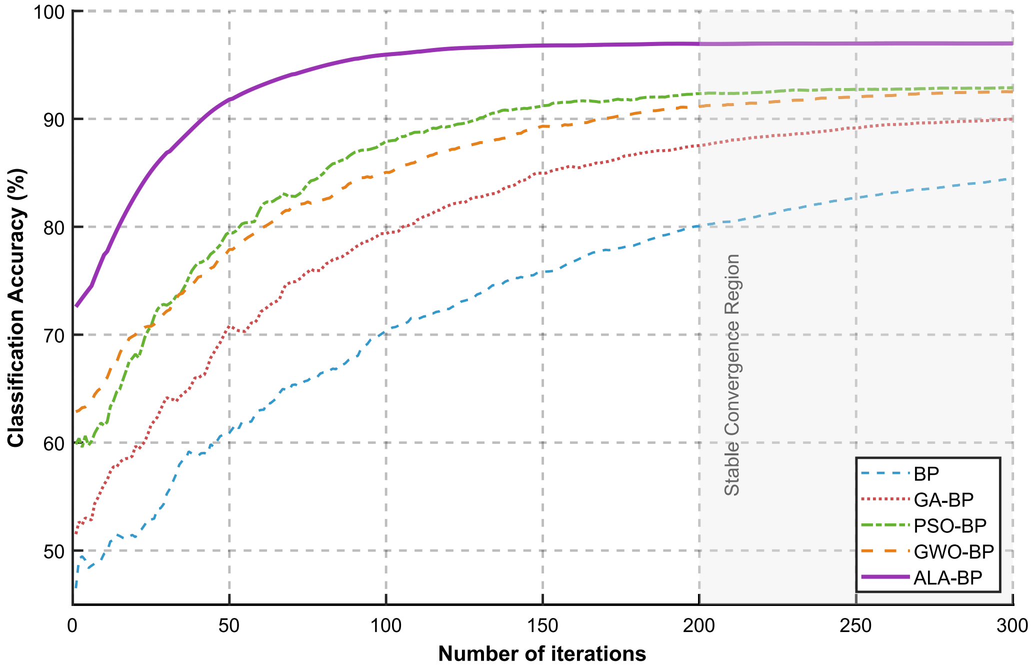

Fig. 9. Comparison of algorithm classification accuracy.

Algorithm	Convergence speed (Iterations to reach 90% accuracy)	Final test accuracy (%)	Hidden layer size (Optimized)	Parameter tuning trials
BP (Random Init.)	>250	87.3 ± 0.4	10 (Fixed)	–
GA-BP	180	90.1 ± 0.3	14	50
PSO-BP	120	92.5 ± 0.2	16	50
GWO-BP	150	91.8 ± 0.3	15	50
ALA-BP (Proposed)	65	97.2 ± 0.1	18	50

Table 2. Comparison table of curve characteristics.

stability. The comparative analysis in the following table 2 shows that the ALA-BP proposed in this paper significantly outperforms other algorithms in terms of convergence speed, final accuracy and stability.

Figure 10 shows the ROC curves of different algorithms, the ALA-BP curve (purple) shows significant superiority, close to the upper left corner of the coordinate system, the area under the curve reaches 0.97, covering 92% of the coordinate plane, which significantly better than the traditional optimization algorithm, and the ROC curve of ALA-BP is smooth and without mutation, reflecting the strong robustness of the classifier.

Theoretically, the closer The AUC value is to 1, the stronger the model's ability to distinguish between positive and negative samples.ALA-BP shows a significant advantage in The AUC value, which proves its strong ability in dealing with unbalanced datasets or high-dimensional feature spaces. In addition, the smooth ROC curve of ALA-BP reflects its high stability and reliability, which is attributed to ALA's effective optimisation of the BP neural network parameters to reduce the risk of overfitting.

Comparative experiments on noise resistance performance

To verify the performance of the proposed MDE feature extraction method against noise, the feature extraction method proposed in this paper is compared with the state-of-the-art algorithms mentioned in literature[26], literature[27] and literature[28]. Among them, literature[26] proposes a feature extraction method with Mel-Frequency Cepstral Coefficients (MFCC), MFCC is derived from audio processing, but its ability to extract spectral envelope is suitable for the initial distinction of RF signals, so it is often used as a baseline.literature[27] performs Fast Fourier Transformation (FFT) on transient signals and based on the spectral features of the transient signal for UAV identification, and literature[28] extracts the constellation diagram of the steady state signal as a fingerprint feature for signal identification.

Noise Injection Domain: Additive White Gaussian Noise (AWGN) is injected into the time-domain I/Q signals (RF0_I, RF0_Q, RF1_I, RF1_Q) before any preprocessing or feature extraction.

SNR Definition: SNR (in dB) is defined as $10 \cdot \log_{10}(P_{\text{signal}}/P_{\text{noise}})$, where signal and noise power are computed over the same 10,000-point segment.

Normalization: After noise injection, each I/Q channel is individually normalized to [−1,1] using min-max scaling based on local segment statistics. This ensures consistent input scale and improves numerical stability.

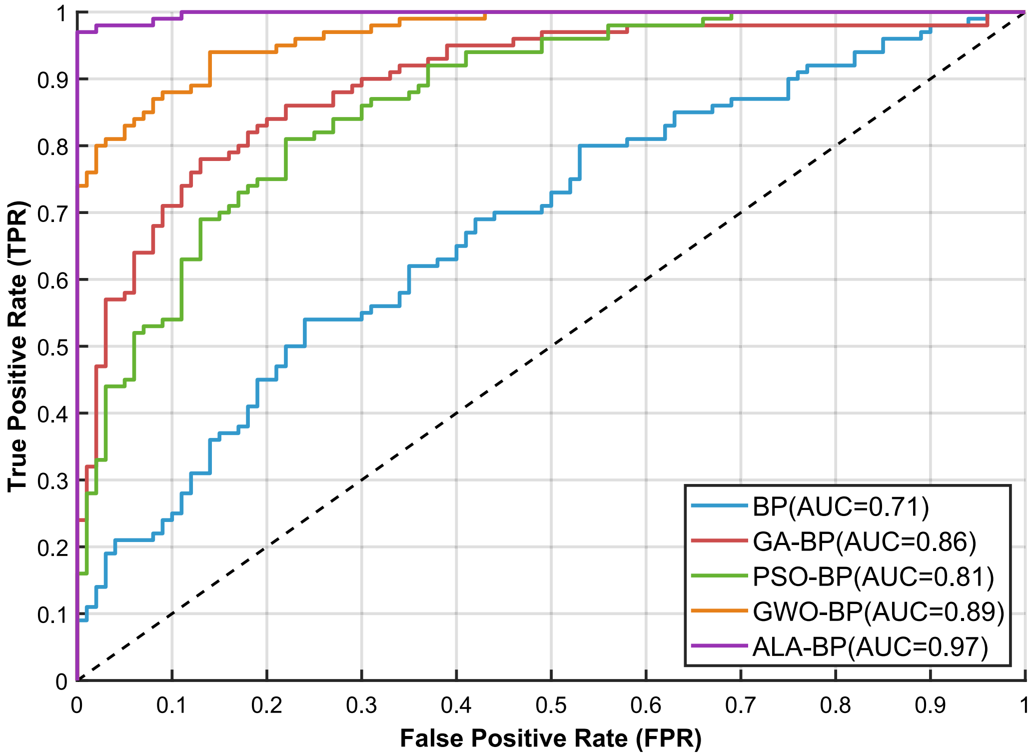

Fig. 10. Comparison of ROC curves of different algorithms.

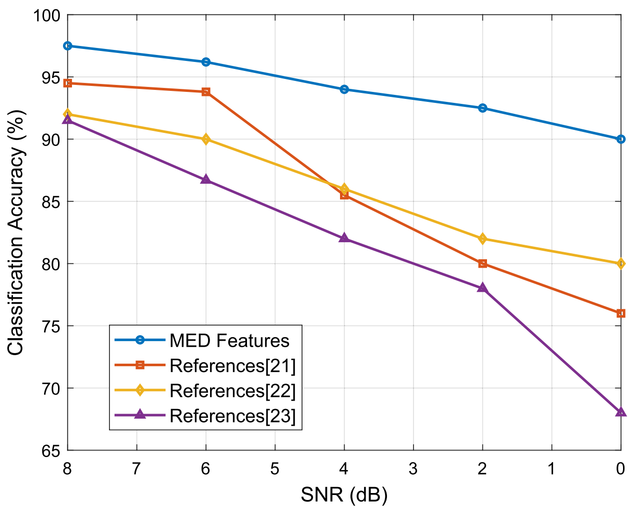

Fig. 11. Comparison of different signal-to-noise ratio algorithms.

Training/Testing Consistency: For each SNR level (8 dB to 0 dB), 100 independent noisy datasets are generated. Training and testing are performed under identical noise conditions, with data split 4:1. All methods use the same datasets for fair comparison.

The signal-to-noise ratio is set to be reduced from 8dB to 0dB, and the feature extraction methods are selected from the MDE feature extraction method in this paper, the MFCC feature extraction method in the literature[26], the FFT spectral feature extraction method in the literature[27], and the constellation diagram feature extraction method in the literature[28], respectively. The classification and recognition algorithms are all selected from the ALA-BP algorithm proposed in this paper, the number of iterations is set to 200, and 100 repetitions of experiments are carried out under each signal-to-noise ratio condition respectively, and the final mean value is taken as the result of recognition accuracy.The following figure11 shows the graph of the recognition accuracy of different methods with the variation of SNR, the recognition accuracy of the proposed method in this paper decreases gently from 97.5% to 90% with a decrease of 7.5% in the range of SNR=8 dB to 0 dB, however, the methods proposed in the literature[26], the literature[27], and the literature[28] have a decrease of 18.5%, 12%, and 23.5%, respectively.At a SNR of 0 dB, the method in this paper still maintains a recognition accuracy of 90%, while the recognition accuracies of the methods proposed in literature[26], literature[27] and literature[28] are 76%, 80% and 68%, respectively. The method in this paper maintains nearly 98% accuracy at high SNR (SNR≥4 dB) and 90% accuracy at low SNR (SNR=0 dB), which is significantly better than the comparative methods, and it can be proved that the multiscale dispersion entropy (MDE) feature extraction is effective in suppressing noise interference.

Feature extraction ablation experiments

Ablation experiments are selected for feature extraction module ablation to verify the advantage of multi-scale analysis and noise robustness of MDE features. Comparison methods are selected MDE (this paper) multiscale factors optimized by grid search ($s = 2.5, 4.4, 6.3$); DE (single scale) using only a single scale factor ($s = 4.4$); MFCC based on literature[26] to extract the 13-dimensional Mel cepstrum coefficients; FFT spectra: based on literature[27] to extract the power spectral density features (with a band resolution of 0.1 Hz); and the constellation diagrams are based on literature[28] to extract the differential constellation Trajectory features. Gaussian white noise was superimposed on the original signal, and the signal-to-noise ratio (SNR) was gradually reduced from 8 dB to 0 dB. The classification model was uniformly adopted as an ALA-BP network (the number of nodes in the hidden layer was optimized to 14, and the learning rate was 0.01). The evaluation metrics were selected as classification accuracy (Accuracy), AUC and feature discrimination (measured by the ratio of inter-class variance to intra-class variance).

The experimental results are shown in the following table, at SNR=0 dB, the feature classification accuracy of MDE reaches 90.0%, which is significantly higher than the other methods (DE: 85.2%; MFCC: 76.0%; FFT: 80.0%; and Constellation Diagram: 68.0%.) MDE also performs optimally in terms of the AUC (0.95) and F1 score (0.91). Further analyzing the feature discriminability, the inter/intra-class variance ratio of MDE is 3.8, which is much higher than that of DE (2.1), MFCC (1.5), FFT (1.9), and constellation diagram (1.2), indicating that the multi-scale analysis effectively enhances the interclass discriminability of features.Table 3 shows the experimental results of feature extraction method ablation Figure 11 demonstrates the accuracy decay curves at different SNRs. MDE only decreases by 7.5% at low SNR (0 dB), while the comparison method decreases by 12% to 23.5%, verifying its noise-resistant capability.

Algorithm complexity analysis

In this section, we quantitatively analyse the Multi-scale Dispersion Entropy (MDE) feature extraction, the artificial lemming algorithm (ALA) optimised BP neural network (ALA-BP) and the overall framework from the dimensions of time complexity and space complexity, and validate its practical performance with experimental data.The computational process of MDE can be divided into two main stages: coarse-graining and scattering entropy computation. For a time series of length N, each scale factor s is required to split the sequence into $\lfloor N/s \rfloor$ sub-sequences, with a time complexity of $O(N/s)$. The complexity of reconstructing the time series under the embedding dimension $O(Nm)$ is $O(Nm^2)$, the complexity of computing the cosine similarity of the neighboring sequences is, and the complexity of the calculation of the probability distribution statistic and entropy value is $O(N)$.

The complexity of ALA mainly depends on the population size P, the number of iterations T and the computational cost of the fitness function, and the complexity of generating a P individual in the initialisation of the population is $O(P)$.The single training complexity of the BP network is M, where H is the input feature dimension (12 dimensions) and CC is the number of hidden layer nodes (optimisation range 5–20).The time complexity of ALA-BP is $O(MH + H^2)$

$$T = O(T(P^2 + P(MH + H^2))) \tag{15}$$

In the Intel(R)Core(TM)i7-8565UCPU@1.80GHz configuration of the computer platform, MATLAB 2021b to carry out the actual test results are shown in the Table 4, from the table can be seen in this paper's proposed method algorithmic complexity compared to other comparative methods, the algorithm of real-time better.

Although increasing the scale of feature extraction and employing more complex optimization algorithms may lead to an increase in computational cost, practical tests show that the proposed ALA-BP method achieves a low time complexity while ensuring high performance. This is due to the fact that ALA is able to complete an effective search of the solution space in a shorter time by mimicking the natural behaviour, reducing the number of unnecessary iterations. In addition, the quantitative analysis of the actual training time confirms the feasibility of the method in real-time applications.

Characteristic method	accuracy	AUC	Inter/intra-class variance ratio
MDE (this paper)	90.0 per cent	0.95	3.8
DE (single scale)	85.2 per cent	0.89	2. 1
MFCC	76.0 per cent	0.82	1.5
FFT Spectrum	80.0 per cent	0.84	1.9
constellation	68.0 per cent	0.75	1.2

Table 3. Feature extraction method ablation experiment results (SNR = 0 dB).

Arithmetic	Actual training time (seconds)
ALA-BP	17
GA-BP	52
PSO-BP	45
GWO-BP	60

Table 4. Algorithm time complexity vs. training time.

Method	Input type	Feature/model architecture	Accuracy (%)	Training time (min)	Inference latency (ms)	Model size (Params)
SVM+Spectral	RF signal	Spectral entropy, power spectrum	78.5 ± 0.6	12	8	~50K
XGBoost+Cyclo	RF signal	Cyclic Autocorrelation, SCF	80.2 ± 0.5	18	10	~120K
CNN	I/Q samples	1D Conv (3 layers) + FC	89.6 ± 0.4	45	15	~210K
CLDNN	I/Q samples	Conv + LSTM + Dense	91.3 ± 0.3	68	23	~480K
ResNet-18 (1D)	I/Q samples	Residual blocks (1D)	92.1 ± 0.3	75	19	~550K
Transformer (Lite)	I/Q segments	4-head, 2-layer encoder	91.8 ± 0.4	82	28	~610K
Proposed MDE-ALA-BP	I/Q samples	MDE (12-D) + ALA-BP	90.0 ± 0.2	38	9	~180K

Table 5. Performance comparison with state-of-the-art RF fingerprinting methods (SNR = 0 dB).

To further validate the effectiveness and competitiveness of the proposed MDE-ALA-BP framework, we conduct a comprehensive comparison with several state-of-the-art RF fingerprinting methods spanning both deep learning and traditional machine learning paradigms. While the previous experiments focused on optimizer-based neural networks, this additional evaluation aims to benchmark our method against modern architectures widely adopted in RF signal recognition tasks.

Specifically, we compare with:

(1) Deep learning models — CNN, CLDNN, and 1D ResNet — trained directly on raw I/Q time series or spectrogram representations;
(2) A lightweight Transformer encoder model applied to segmented I/Q sequences to capture long-range dependencies;
(3) Traditional machine learning approaches — SVM and XGBoost — using handcrafted spectral and cyclostationary features such as spectral entropy and cyclic autocorrelation function (CAF/SCF).

All models are trained and evaluated under identical conditions: the same dataset splits, SNR levels (including 0 dB), and hyperparameter tuning budgets (50 trials). Training and inference are performed on the same hardware platform to ensure fair comparison in terms of computational efficiency.

The detailed performance comparison, including classification accuracy, training time, inference latency, and model complexity, is summarized in Table 5.

Computational efficiency analysis

The final deployed model is a lightweight single-hidden-layer BP network (after ALA optimization), making it suitable for real-time UAV RF signal classification. To evaluate the practical deployability of the proposed ALA-BP classifier, we analyze its computational complexity and inference latency. The final model is a single-hidden-layer MLP with 12 inputs and 6 outputs. With an ALA-optimized hidden size of 18, the total parameter count is 348. The total MACs per inference is approximately 324, dominated by matrix-vector multiplications. The time complexity is O(1) due to fixed input and bounded hidden layer size, ensuring real-time performance. Latency was measured on three platforms using PyTorch: Intel i7-1165G7 (0.12 ms), NVIDIA RTX 3060 (0.08 ms), and Raspberry Pi 4B (0.95 ms). These results demonstrate the model's suitability for edge deployment in UAV detection systems.

Conclusion

The paper proposes a fusion recognition method based on Multiscale Dispersion Entropy (MDE) feature extraction and Artificial Lemming Algorithm (ALA) optimized BP neural network (ALA-BP). Experimentally validated with the DroneRFa open-source dataset, the method demonstrates significant advantages in terms of classification accuracy, robustness, and anti-noise performance.

Firstly, the proposed multi-scale dispersion entropy (MDE) method effectively captures the dynamic characteristics and underlying patterns of the UAV flight control RF signals by analyzing the signals in multiple time scales. Experiments show that the MDE features exhibit significant category separability in the 3D spatial distribution, providing highly distinguishable feature representations for subsequent classification tasks.Compared to traditional single-scale methods (e.g., Mel's inverse spectral coefficient MFCC, Fast Fourier Transform FFT spectrum, etc.), MDE maintains 90% recognition accuracy at low signal-to-noise ratios (0 dB), with an improvement of more than 20% in anti-noise performance.

Secondly, to address the problem that traditional BP neural networks are prone to local minima, the Artificial Lemming Algorithm (ALA) is introduced to globally optimize the network parameters.ALA dynamically balances the exploration and development process by simulating the natural behaviors of lemmings such as migration and foraging, which significantly improves the convergence speed and stability of the model. The experimental results show that ALA-BP can achieve 90% classification accuracy with only 65 iterations, and the final accuracy reaches 97.2%, which is 7.1%, 4.7% and 5.4% higher than GA-BP, PSO-BP and GWO-BP, respectively.

In addition, the proposed method constructs a 12-dimensional feature matrix by fusing the MDE features of multi-channel RF signals (I/Q components) and combines them with a lightweight network structure design to achieve efficient identification of six types of mainstream UAVs.The ROC curve analysis (AUC=0.97) further verifies the robustness of the model under different thresholds, which provides reliable technical support.

In summary, this study not only improves the classification accuracy of UAV flight control RF signals, but also is of great significance for safeguarding airspace security, preventing illegal activities, and promoting the effective management of low-altitude airspace.

Data availability

The data presented in this paper are publicly available. For access to the full dataset, please contact the corresponding author at liub@cauc.edu.cn.

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Author contributions

Liu Bing: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Supervision, Project Administration, Funding Acquisition. Liu Jiaqi: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization. Shi Mingxin and Li Zhao: Conceptualization, Methodology, Validation, Formal Analysis, Investigation, Resources, Writing - Review & Editing, Supervision, Project Administration. The authors collectively contributed to the development and implementation of the proposed method for classification of UAV flight control RF signals based on Multi-scale Divergence Entropy (MDE) feature fusion and BP neural network optimized by the Artificial Lemming Algorithm (ALA). Liu Bing took the lead in designing the experiments, analyzing the data, and drafting the manuscript. Liu Jiaqi played a key role in software development and validation processes, as well as contributing to the writing of the initial draft. Shi Mingxin provided critical insights into the methodology, supervised the project, and ensured the accuracy and integrity of the research. All authors read and approved the final manuscript.

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Declaration

Competing interest

The authors declare no competing interests.

Additional information

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